

The Chain Rule & Directional Derivatives



The multivariable chain rule

- 1 Let $z = x^2 + y^2$, $x = \cos t$, $y = \sin t$. Find $\frac{dz}{dt}$.
 - 2 Let $w = xy + z^2$, $x = t$, $y = t^2$, $z = t^3$. Find $\frac{dw}{dt}$ at $t = 1$.
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Chain rule with two independent variables

- 3 Let $z = f(x, y)$ where $x = s + t$, $y = s - t$. Express $\frac{\partial z}{\partial s}$ in terms of $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.
 - 4 For $z = x^2y$, $x = s + t$, $y = s - t$, find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$ at $s = 2$, $t = 1$.
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Gradient and directional derivative

- 5 Find the gradient of $f(x, y) = x^2y^3$ at $(1, 2)$.
 - 6 Find the directional derivative of $f(x, y) = x^2y$ at $(2, 1)$ in the direction of $\mathbf{v} = \langle 1, 1 \rangle$.
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The gradient points in the direction of steepest ascent

- 7 For $f(x, y) = 3x^2 - y^2$ at $(1, 2)$, find the direction of steepest ascent and the maximum rate of increase.
 - 8 In what direction from $(1, 1)$ does $f(x, y) = xy$ decrease most rapidly, and at what rate?
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Application of the gradient

- 9 The temperature on a plate is $T(x, y) = 100 - x^2 - 2y^2$. A bug at $(2, 1)$ wants to cool off fastest. In which direction should it move?
 - 10 Find the rate of change of $T(x, y) = 100 - x^2 - 2y^2$ at $(2, 1)$ in the direction of the unit vector $\langle \frac{3}{5}, \frac{4}{5} \rangle$.
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More chain-rule practice

- 11 Let $z = x^2 - y^2$, $x = e^t$, $y = e^{-t}$. Find $\frac{dz}{dt}$.
 - 12 Let $w = x^2 + y^2 + z^2$, $x = t$, $y = t^2$, $z = t^3$. Find $\frac{dw}{dt}$ at $t = 1$.
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More gradient and directional derivative practice

13 Find the gradient of $f(x, y) = e^{xy}$ at $(1, 0)$.

14 Find the directional derivative of $f(x, y) = x^2 + y^2$ at $(3, 4)$ in the direction of $\mathbf{v} = \langle -4, 3 \rangle$.

The tangent line to a level curve

15 Using that ∇f is normal to the level curve of f , find the tangent line to $x^2 + y^2 = 25$ at $(3, 4)$.

The chain rule with three intermediate variables

16 For $w = f(x, y, z)$ with $x = g(s, t)$, $y = h(s, t)$, $z = k(s, t)$, write the chain rule formula for $\frac{\partial w}{\partial t}$.

One more gradient computation

17 Find the gradient of $f(x, y) = \ln(x^2 + y^2)$ at $(1, 2)$.

18 Is the direction $\langle 1, -1 \rangle$ a direction of increase or decrease for $f(x, y) = x + y$ at any point?

A closing chain-rule problem

19 Let $z = \sin(x) \cos(y)$, $x = t^2$, $y = t$. Find $\frac{dz}{dt}$ at $t = 0$.

A final directional derivative problem

20 Find the directional derivative of $f(x, y) = x^2 - y^2$ at $(2, -1)$ in the direction from $(2, -1)$ toward $(5, 3)$.

Implicit differentiation via the chain rule

21 If $x^2y + y^3 = 8$ defines y implicitly as a function of x , use $\frac{dy}{dx} = -\frac{F_x}{F_y}$ (for $F(x, y) = x^2y + y^3 - 8$) to find $\frac{dy}{dx}$.

22 Evaluate $\frac{dy}{dx}$ from the previous question at the point $(0, 2)$ (check first: $0^2(2) + 2^3 = 8$ ✓, so this point is on the curve).

A last pair of gradient problems

23 Find the gradient of $f(x, y, z) = x^2 + y^2 + z^2$ at $(1, 2, 2)$, and find its magnitude.

24 For $f(x, y) = x^2y$, find the maximum rate of change of f at $(3, 1)$ and the direction in which it occurs.

Chain rule with a three-step composition

25 Let $z = x^2y$, $x = r \cos \theta$, $y = r \sin \theta$ (polar substitution). Find $\frac{\partial z}{\partial r}$.

26 Using the substitution above, evaluate $\frac{\partial z}{\partial r}$ at $r = 2$, $\theta = 0$.

A last directional-derivative setup

27 Find the directional derivative of $f(x, y) = xe^y$ at $(1, 0)$ in the direction $\theta = \frac{\pi}{3}$ from the positive x -axis.